# William J Vianese

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UK Work Authorization: High Potential Individual (HPI) Visa | No sponsorship required

## EDUCATION

**WASHINGTON UNIVERSITY IN ST. LOUIS** | St. Louis, MO **May 2026**

*Bachelor of Arts*

Majors in Math & Computer Science and Environmental Analysis

- GPA: 3.84/4.00 (First-Class Honors equivalent); College Honors; Dean's List (Spring 2023, Fall 2023, Fall 2024, Fall 2025)

Relevant Coursework

- Data Science, Data Mining, Artificial Intelligence, Analysis of Algorithms, Algorithms for Nonlinear Optimization, Data Structures, Dynamics & Computation, Financial Mathematics, Probability & Statistics, Matrix Algebra, and Applied Math

## TECHNICAL SKILLS

- *Languages:* Python, SQL, Java, R
- *Libraries:* Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn
- *Tools:* GitHub, Microsoft Excel, Claude Code, Codex

## PROJECTS

*Ferguson Housing Abandonment Risk Map*

- Built an interactive geospatial web application to visualize housing risk for 8,800+ parcels with layered map views, filters, and address search for exploratory analysis
- Used CLIP vision embeddings on Google Street View imagery to classify parcels as occupied, abandoned, or vacant
- Trained a logistic regression model on labeled county records to predict abandonment risk for previously unclassified parcels
- Combined computer vision and statistical model outputs into a unified parcel-level risk score and displayed key predictive variables to support interpretation of neighborhood-level risk patterns

*Real-Time Multiplayer Monopoly Game*

- Designed and built a real-time, full-stack multiplayer game platform with server-side game logic and state synchronization
- Built event-driven turn logic using WebSockets to support concurrent players, auctions, trading, and game state updates
- Implemented persistence and session management to support reconnects and multi-game concurrency
- Deployed the application to a cloud hosting platform and tested with live users

*Melbourne Housing Price Prediction*

- Built and compared linear regression, k-nearest neighbors, decision tree, and random forest models to predict housing prices
- Preprocessed mixed numerical and categorical data using imputation, one-hot encoding, scaling, and train-test pipelines
- Applied cross-validation and hyperparameter tuning, evaluating models using RMSE and R-squared
- Selected features and compared models before and after feature engineering to evaluate the impact on predictive performance

## EXPERIENCE

**CAMP EMERSON** | Hinsdale, MA **May 2022 – August 2025**

*Roles Held: Data & Systems Manager, Program Director, Wilderness Director, and Counselor*

- Progressed across four roles to own and optimize core operational systems supporting 285+ users and 80+ staff, spanning enrollment, scheduling, transportation, billing, and compliance
- Designed and implemented automated workflows and process improvements that increased data consistency, reduced manual operations, and improved system reliability
- Performed data analysis and reporting on enrollment trends, logistical constraints, financial activity, and staffing needs to support leadership decisions

## ACTIVITIES

**WASHINGTON UNIVERSITY ROWING TEAM** | St. Louis, MO **August 2022 - May 2026**

*Coach and Coxswain*

- Lead workouts, dieting, and recovery in conjunction with motivating and organizing the rowers
- Employ critical and quick-thinking skills to practice effectively on the water and land
- Leverage strong interpersonal skills to promote team cohesion and expand motivation

## AWARDS AND RECOGNITIONS

- Selected as ACRA Third Team All-American Coxswain 2024
- Selected as ACRA All-Regional Coxswain in the Plains Region 2023, 2024